# CS 258 Parallel Day

**Introduction to Parallel Computing: Data Parallelism**

**Activity: Flag Maker**

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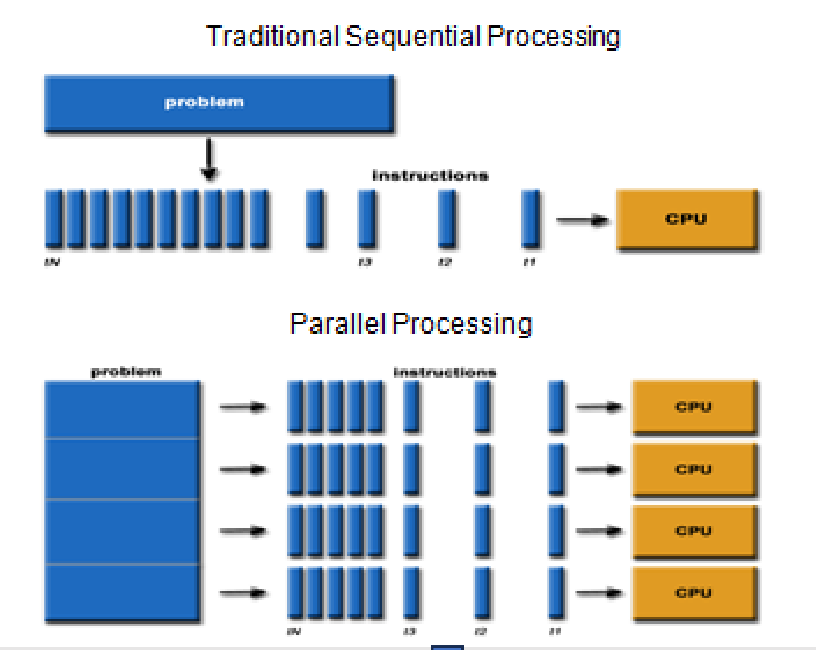
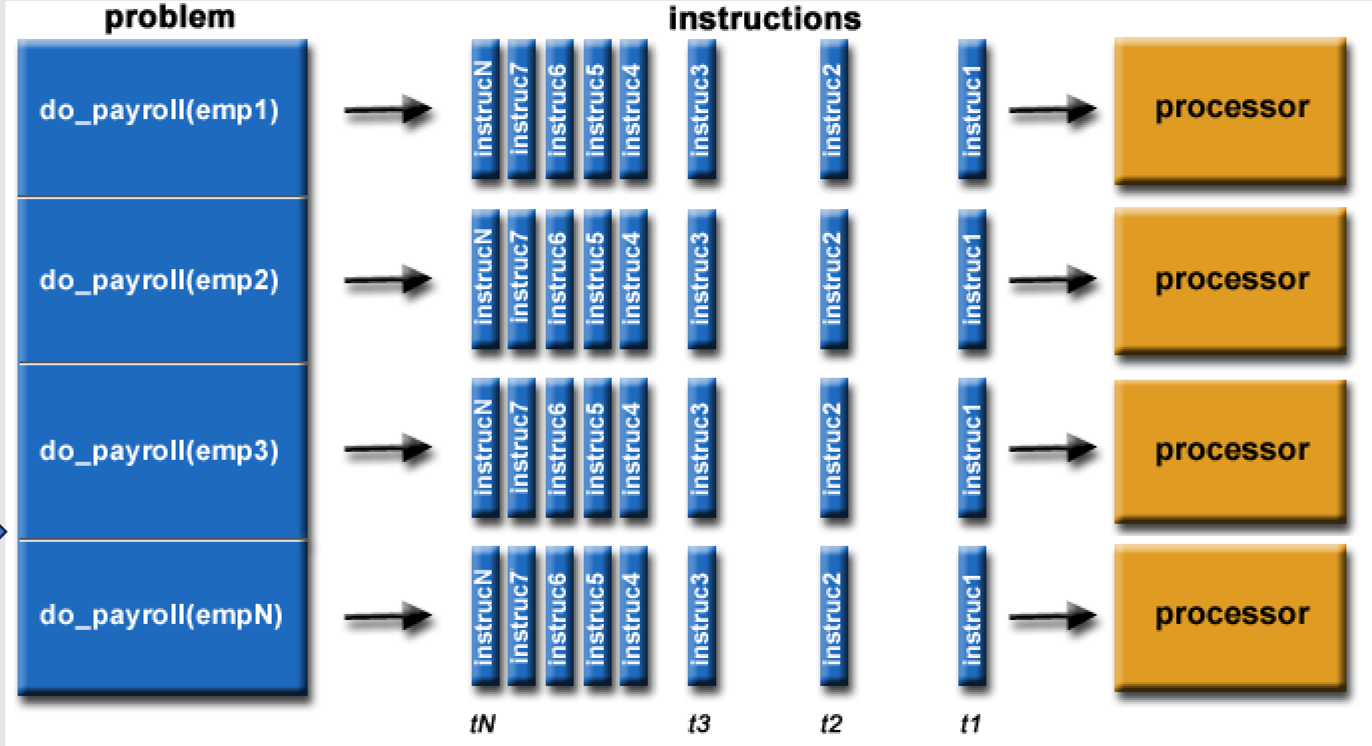
STEP 1: Pre-Activity Quiz to Assess Pre-Knowledge

Link on Blackboard

# What is Parallel Computing?



# What is Parallel Computing?

- Digging Holes: https://youtu.be/bun_WSB9iRw?si=TOVqmlKx6OEmXmnd
- More video for later: https://www.youtube.com/watch?v=GXrI3oJ3Eak
- Detailed explanations:
  - https://computing.llnl.gov/tutorials/parallel_comp/#Abstract
  - https://cse.unl.edu/~cbourke/PDCatUNL/module-parallel.html

## Glossary of PDC Terms for our Flag Maker Activity

- Parallelism (Data/Task)
  - In our activity we are using **data parallelism**
- Sequential computing
- Parallel computing
- Multicore
- Multiprocessor
- Synchronization
- Scheduler
- Race condition

# STEP 2: Flag Maker Activity

Img from: https://en.wikipedia.org/wiki/Mauritius

- You'll be pretending to be the computer drawing the flag of Mauritius (on the left).
- You'll use paper and coloring markers but otherwise follow how a computer might draw it.
- You must fill in the squares completely.
- You must color one square at a time.
  Do not color in multiple squares at one time.

# FlagMaker Activity

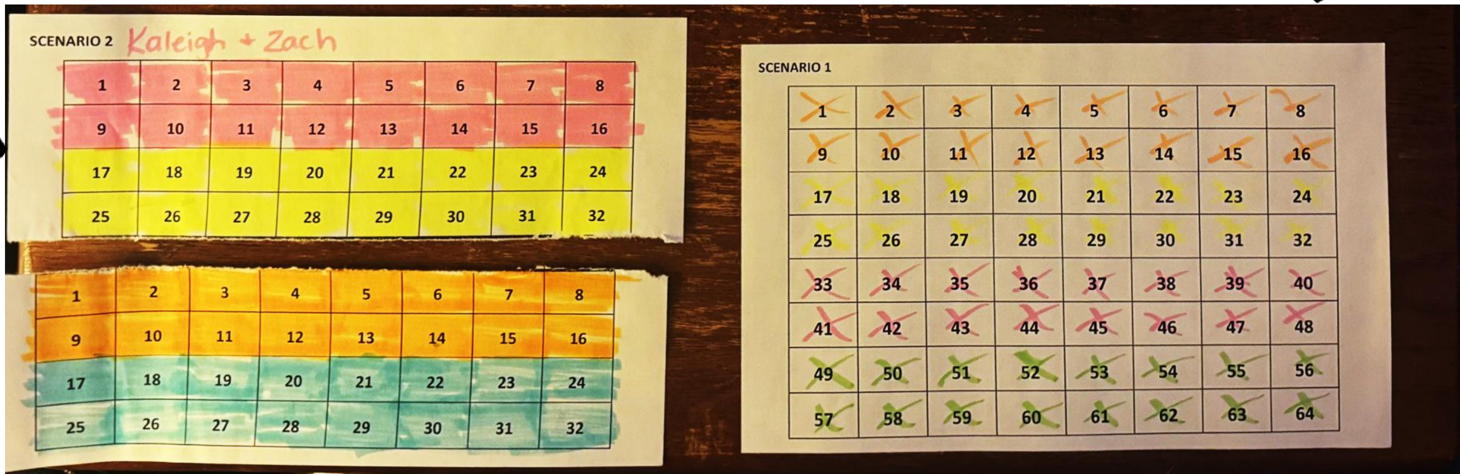
You must **color one square at a time**.

Do not color in multiple squares at one time.


**CORRECT!**

**WRONG!**

**Fill in the squares**




**WRONG!**

**Don't tear the paper!**

# Scenario 1: A Single Processor

1. **The Processor**: colors first stripe (cell by cell), second stripe (cell by cell), third stripe (cell by cell), and then fourth stripe (cell by cell)
2. **Timer**: Times the Processor
   - Write the time on your flag sheet

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| P1 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
| | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 |
| | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 |
| | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 |
| | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 |
| | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 |

**Remember: squares must be mostly colored in & you can only color one square at a time, left to right, top to bottom!**

# Scenario 2: Two Processors Collaborating

**Three people:**

1. **Processor 1**: Colors red and blue stripes (cell by cell)
2. **Processor 2**: Colors yellow and green stripes (cell by cell)
3. **Timer**: Times them; stop when **all** cells are colored. The overall parallel time is calculated based on the slowest processor.
   - Write the time on your flag sheet

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| P1 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
| | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 |
| | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 |
| P2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
| | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 |
| | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 |

**Remember:** **squares must be mostly colored in & you can only color one square at a time!**

# Scenario 2: Two Processors Collaborating

## Let's Discuss

1. How does the addition of a second processor impact the overall completion time?

2. What potential problems could arise if both processors were given overlapping tasks such as coloring the same stripes?

# Scenario 3: Four Processors Collaborating

Five people:

1. Processors 1-4: Each assigned to one color, which they color (cell by cell)
2. Timer: Times them; stop when **all** cells are colored. The overall parallel time is calculated based on the slowest processor.

- Write the time on your flag sheet

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| **P1** | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
| **P2** | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
| **P3** | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
| **P4** | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 |

**Remember:** squares must be mostly colored in & you can only color one square at a time!

# Scenario 3: Four Processors Collaborating

## Let’s Discuss

1. How does using four processors further improve efficiency, and what challenges might this bring?

2. What would happen if one processor finished its task significantly earlier than the others?

# Scenario 4: Four Processors Collaborating a Different Way (Column Assignments)

Five people:

1. Processors 1-4: Each assigned a strip of two columns (first takes left 2 columns, second takes next two columns, etc.) Processors colors these the correct color cell by cell.
2. Timer: Times them; stop when **all** cells are colored. Times the slowest processor

- Write the time on your flag sheet

| P1 | | P2 | | P3 | | P4 | |
|---|---|---|---|---|---|---|---|
| 1 | 9 | 1 | 9 | 1 | 9 | 1 | 9 |
| 2 | 10 | 2 | 10 | 2 | 10 | 2 | 10 |
| 3 | 11 | 3 | 11 | 3 | 11 | 3 | 11 |
| 4 | 12 | 4 | 12 | 4 | 12 | 4 | 12 |
| 5 | 13 | 5 | 13 | 5 | 13 | 5 | 13 |
| 6 | 14 | 6 | 14 | 6 | 14 | 6 | 14 |
| 7 | 15 | 7 | 15 | 7 | 15 | 7 | 15 |
| 8 | 16 | 8 | 16 | 8 | 16 | 8 | 16 |

**Remember:** squares must be mostly colored in & you can only color one square at a time!

# Scenario 4: Four Processors Collaborating a Different Way (Column Assignments)

## Let's Discuss

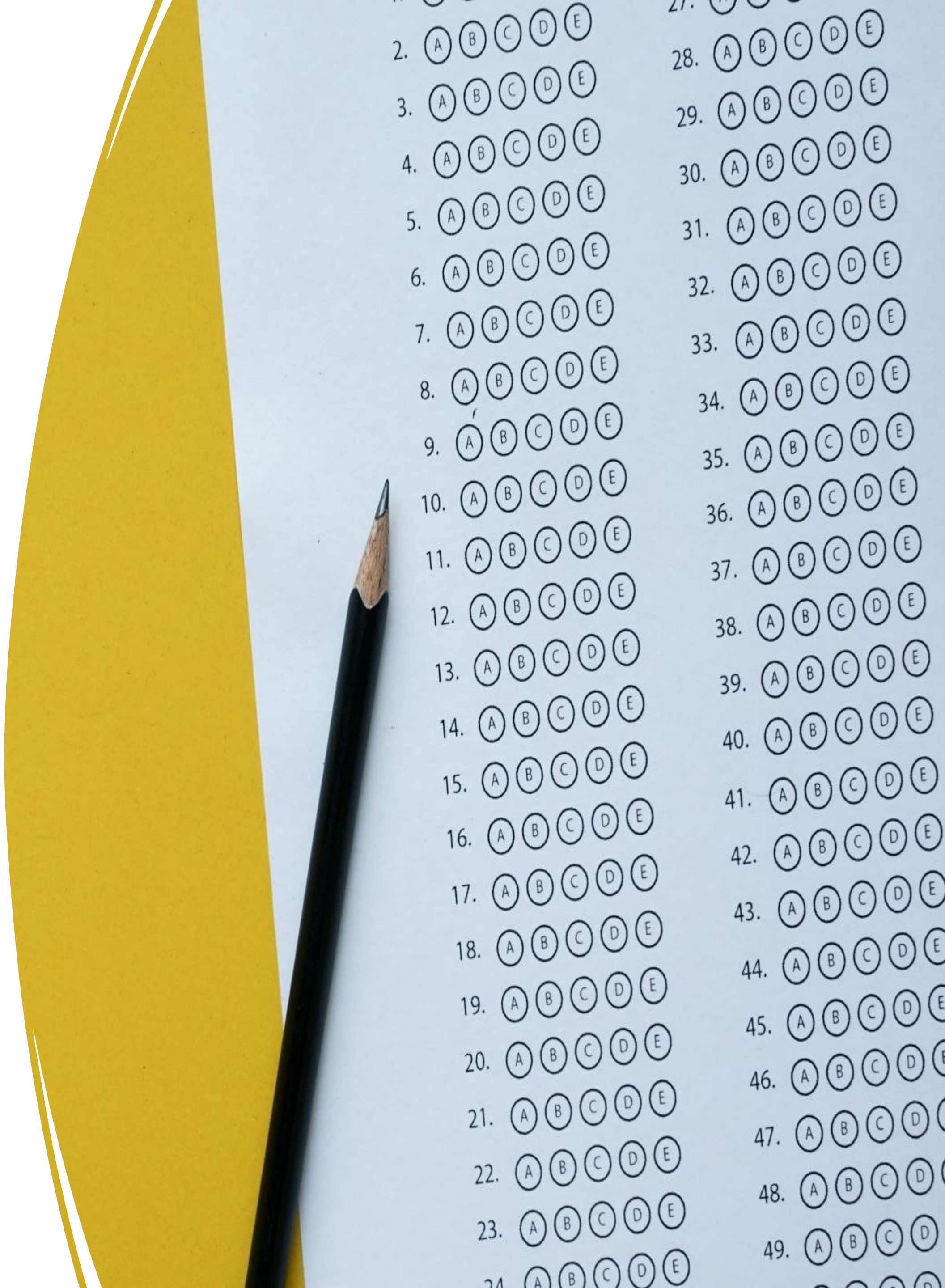
1. How does assigning processors by columns rather than stripes impact performance and potential errors?

2. How does this scenario reflect real-world issues in distributed systems?

# STEP 3: Post-Activity Quiz to Assess Post-Knowledge

- Please take this post-test quiz:

https://forms.gle/NYiKx2RtC7Jq9muT7



# STEP 4: Post-Activity Student Engagement Survey

- Please click on the following link to take the survey

https://usisurvey.az1.qualtrics.com/jfe/form/SV_4UQgARfhKOefDQG

# Thank You!